MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MODEL QUESTION PAPER - THEORY

STATISTICS

MG5DSCSTA301 – SAMPLING TECHNIQUES

Duration: 2 Hours **Maximum: 70 Marks**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer questions

Answer any ten questions. Each question carries **3 marks**.

1. What do you mean by non-probability sampling ? [U] [1]
2. What is Quota sampling? [U] [1]
3. What is the probability of getting a specified sample in SRSWOR? [K][2]
4. Describe the procedure of preparing slips in the Lottery Method. What precautions should be taken? [K][2]
5. What is meant by an unbiased estimator of population mean under SRSWR? [U][2]
6. What is sampling fraction? [U][2]
7. Define stratum weight for the hth stratum under stratified random sampling.[K][3]
8. What is proportional allocation? [U][3]
9. Explain how you will draw a linear systematic sample.[U][4]
10. Explain circular systematic sampling.[K][4]
11. What are the advantages of systematic sampling? [K][4]
12. What do you mean by linear trend in population values? [K][4]

Part B

Short Essay questions

Answer any four questions. Each question carries **7 marks.**

13. Explain the merits and demerits of Probability Sampling ? [K][1]

14. Explain how you would select a random sample of 11 households from a list of 112 households in a village using random number table method. [A][2]
15. Explain complete procedure of selecting random sample with replacement.[K][2]
16. What is Strata? What are the principles of stratification?[U][3]
17. A flood affected district in Kerala is divided into three zones. The number of villages in each zone is given by 440, 405, and 100 respectively. In order to estimate the total number of houses affected by flood in the district, the number of villages selected from each zone are 25,15 and 8 respectively. If total number of houses affected in the sampled villages are 750, 600, 344 respectively, estimate the total number of houses affected by flood in the district. [A][3]
18. Give an unbiased estimator of population total based on cluster sampling where the clusters are of equal Size. [U][5]

Part C

Essay questions

Answer any one question. Each question carries **12 marks**.

19. Explain in detail the various types of Probability Sampling with suitable examples ? [U][1]

20. Carryout a comparison between systematic sampling and stratified random sampling in the case of a population with linear trend among observations.[U][4]